Unifying Theory of Dynamic Aether Mechanics: Strong and Weak Nuclear Forces

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Abstract

Within the Dynamic Aether Mechanics framework, the strong force arises from compressed Aether bridges—physical strands of quantum superfluid compressed to $\sim 7 \times 10^7$ times ambient density, connecting quark vortex endpoints. This paper derives confinement (bridge snapping at ~1.2 fm creates new quark–antiquark pairs), asymptotic freedom (bridges are "all core" with compactness $d/\xi \approx$ 1–2), hadron mass (predominantly bridge compression energy), and the deconfinement phase transition at $T \approx 170$ MeV. Quantum fluctuations of the bridge reproduce the Lüscher term $-\pi/(12\,r)$, predict Schwinger pair-creation rates that naturally separate chiral from potential-model QCD at a crossover mass of ~250 MeV, and explain the nucleon sigma term. The weak force is identified as vortex topology transitions. A unified summary table relates all four fundamental forces through distinct Aether processes.

This paper is part of a series presenting the Unifying Theory of Dynamic Aether Mechanics. For the foundational concepts of Aether binding, the three independent properties (density, pressure, temperature), and the unification of force constants, see Paper 1: General Overview [13].

1. The Strong Force: Compressed Aether Bridges

The three forces discussed so far—magnetism, the electric field, and gravity—all arise from *unbound* Aether: organized flow patterns, coherent axial-oscillation wave emission from bound core rings, and the residual asymmetry of transient binding. But the theory's central mechanism—the ability of Aether to *bind* into matter—has not yet been applied to the strongest force in nature.

The strong nuclear force binds quarks into protons and neutrons, and protons and neutrons into nuclei. Its defining features are well established experimentally: confinement (free quarks are never observed), asymptotic freedom (quarks behave as free particles at short distances), a force that is approximately constant with distance (the "string tension" $\sigma \approx 1$ GeV/fm), and spontaneous quark–antiquark pair creation when quarks are pulled apart. In the Standard Model, these properties emerge from quantum chromodynamics (QCD), a gauge theory with the symmetry group SU(3).

In the Aether framework, the strong force has a direct physical interpretation: it is the elastic energy stored in a *persistent bridge of compressed Aether* connecting quark vortex endpoints.

1.1 The Binding Bridge

When Aether binds permanently (as described in Section 2), the bound material no longer flows or exerts pressure as a fluid—it is locked into mass. In the case of the electron, binding produces a closed vortex ring whose stability rests on circulation quantization. But quark vortex endpoints can also anchor an open structure: a strand of highly compressed Aether connecting two endpoints, held in its compressed state by the topological boundary conditions the vortices impose.

In QCD, this structure is known as a *color flux tube*—a narrow tube of chromodynamic field energy connecting a quark to an antiquark (in a meson) or linking three quarks at a Y-shaped junction (in a baryon). Lattice QCD calculations confirm that the chromodynamic field is concentrated in such tubes rather than spreading through all space. In the Aether model, the flux tube is not an abstract field configuration but a physical object: a strand of compressed Aether with a definite cross-sectional area, energy density, and material properties.

The energy of the bridge is proportional to its length:

$$E_{\text{bridge}} = \sigma \cdot L \tag{1}$$

where σ is the string tension (energy per unit length) and L is the bridge length. This linear energy–distance relationship produces a *constant* restoring force:

$$F_{\text{strong}} = \frac{dE}{dL} = \sigma \approx 1 \text{ GeV/fm} \tag{2}$$

A force that does not weaken with distance is the signature of a material connection rather than a field that spreads through space. The electric force weakens as $1/r^2$ because the coherent axial-oscillation wave intensity dilutes over a sphere. The strong force does not weaken because the bridge does not dilute—it is a one-dimensional material strand, not a three-dimensional wave field.

1.2 Confinement

The confinement of quarks—the experimental fact that no free quark has ever been observed—follows directly from the bridge model. When energy is supplied to separate two quarks, the bridge stretches, storing energy σL. When the stored energy reaches the threshold for creating a new quark–antiquark pair—including both the rest mass and the kinetic energy of the newly created quarks within the resulting mesons, totalling ~1 GeV—the bridge snaps. Each broken end immediately forms a new vortex endpoint, producing two shorter bridges instead of one long one. The result is two mesons (quark–antiquark pairs), not two free quarks. Lattice QCD simulations confirm that this string-breaking transition occurs at a quark separation of $r_{\text{break}} \approx 1.2$ fm [4, 5]—consistent with $\sigma \times r_{\text{break}} \approx 1.2$ GeV after accounting for the short-distance Coulomb contribution to the static potential.

This is exactly the mechanism of permanent Aether binding applied to a one-dimensional geometry: when the energy density in the bridge exceeds the Dynamic Casimir Effect threshold, new bound matter forms at the break point. The same DCE threshold that governs electron creation governs the snap point of the bridge.

Confinement is therefore not a mysterious emergent property of a non-Abelian gauge theory—it is the straightforward consequence of stretching a material strand until it breaks and forms new endpoints.

1.3 Asymptotic Freedom

At very short distances, the strong coupling constant α_s decreases—quarks inside a hadron interact more weakly when close together. This "asymptotic freedom" is one of the most counterintuitive features of QCD.

In the bridge model, asymptotic freedom has a geometric explanation rooted in the bridge's compactness. The bridge diameter $d_{\text{bridge}} \approx 0.30$–$0.35$ fm is only 1–2 times the healing length $\xi \approx 0.17$–0.28 fm (Section 4.8). This means the bridge has no extended "bulk interior"—the entire cross-section is within the healing-length transition region, where the condensate density rises smoothly from the ambient value to its peak and back.

When two quarks are separated by less than the bridge diameter ($L \lesssim d_{\text{bridge}} \sim 1$–$2\,\xi$), the compressed structure between them has not fully formed: the condensate perturbation is still developing over the $\sim \xi$ recovery length on each side. The quarks sit inside a partially-formed, low-amplitude perturbation rather than a fully-compressed strand, so the effective restoring force is weak. As the quarks separate and L exceeds d_{bridge}, the bridge becomes a well-defined one-dimensional strand with full string tension σ, and the confining force reaches its asymptotic constant value.

This predicts a specific crossover scale: the coupling should strengthen rapidly as the quark separation passes through $\sim\xi \approx 0.2$ fm, consistent with lattice QCD measurements showing the quark-antiquark potential transitioning from approximately Coulombic ($V \propto 1/r$) at short range to linear ($V = \sigma r$) at long range, with the crossover at ~0.2–0.5 fm. The crossover scale is not a free parameter—it is determined by the healing length ξ, which is itself fixed by the Aether quantum mass m_A.

Asymptotic freedom in this picture is not a running coupling constant in an abstract renormalization group, but a geometric transition: from a region too small for the condensate to form a coherent compressed structure (weak coupling) to a fully-formed bridge (constant confining force).

1.4 Gluon Self-Interaction

In quantum electrodynamics, photons do not interact with each other—electromagnetic waves in the Aether are linear (to the precision confirmed by experiment). In QCD, gluons interact with each other, producing the non-Abelian structure that distinguishes the strong force from electromagnetism.

In the bridge model, this distinction is natural. Photons are compression waves in *unbound* Aether, which is experimentally confirmed to be perfectly linear (Hookean) in its pressure–density response. Gluons are excitations of the *compressed* bridge material—density waves, vibrational modes, and distortions traveling along the compressed strand. Excitations in a highly compressed medium are inherently nonlinear: phonons scatter off each other, solitons interact, and density waves couple. The gluon self-interaction is not a postulate but the expected behavior of disturbances propagating through a compressed material bridge.

This also explains why the strong force is non-Abelian while electromagnetism is Abelian: the unbound Aether is a simple elastic medium (linear, Abelian), while the compressed bridge interior is a dense, nonlinear system (non-Abelian). The mathematical difference between the two forces reflects the physical difference between wave propagation in a free medium and wave propagation in a material strand.

1.5 The Origin of Hadron Mass

The proton has a mass of 938 MeV/c^2, but the three valence quarks inside it (two up quarks at ~2.2 MeV and one down quark at ~4.7 MeV) account for only ~9 MeV—less than 1% of the total. The remaining 99% is attributed in QCD to "gluon field energy" and "quark kinetic energy."

In the bridge model, this mass has a concrete physical identity: it is the *elastic compression energy stored in the bridges*. The bridge material is Aether compressed to $\sim 7 \times 10^7$ times ambient density, and this compression energy is mass ($E = mc^2$). The proton's mass is predominantly the mass of the Y-shaped bridge connecting its three quark endpoints, not the mass of the endpoints themselves.

This reinterpretation makes a quantitative prediction: the proton mass should be calculable from the string tension σ, the bridge geometry (Y-shaped, with three arms of length ~0.3 fm meeting at a junction), and the endpoint (quark) masses:

$$m_p c^2 \approx 3 \times \sigma \times L_{\text{arm}} + (2m_u + m_d)\, c^2 + E_{\text{junction}} \tag{3}$$

With $\sigma \approx 1$ GeV/fm and $L_{\text{arm}} \approx 0.3$ fm, the bridge contribution is ~900 MeV, which is the correct order of magnitude. The junction energy E_{junction} represents the binding energy at the point where three bridges meet.

1.6 Color as Bridge Orientation

In QCD, quarks carry one of three "color" charges (red, green, blue), and observable hadrons must be color-neutral. In the Aether framework, the three colors correspond to the *three independent spatial orientations* of the binding bridge.

A vortex line in three spatial dimensions has its circulation confined to a plane. There are three independent planes (conventionally xy, xz, yz), and a binding bridge inherits the orientation of the vortex circulation at its endpoints. The three colors are these three orientations:

- A bridge in the xy-plane carries "red" color.
- A bridge in the xz-plane carries "green" color.
- A bridge in the yz-plane carries "blue" color.

Color neutrality then has a geometric meaning: a baryon contains three bridges spanning all three spatial planes, producing a composite structure that is rotationally invariant—it has no preferred orientation. A meson contains a bridge and its conjugate (a quark and an antiquark with opposite color orientations in the same plane), which is also rotationally neutral in that plane.

The step from three orientations to SU(3) gauge symmetry requires the quantum nature of the bridges. Classically, rotations among three planes give SO(3), which has only 3 generators—far fewer than the 8 generators of SU(3). But bridges in a quantum superfluid are quantum objects that can exist in coherent superpositions of orientations:

$$|\psi_{\text{color}}\rangle = \alpha\,|xy\rangle + \beta\,|xz\rangle + \gamma\,|yz\rangle \tag{4}$$

where α, β, γ are complex amplitudes. Unitary transformations on this three-component complex vector space form U(3), and after removing the overall phase, SU(3)—which has exactly 8 generators (the Gell-Mann matrices). In an isotropic medium all three orientations are energetically equivalent, guaranteeing at minimum SO(3) invariance. If the dynamics is also invariant under relative complex-phase rotations between orientations—so that $|xy\rangle + |xz\rangle$ and $|xy\rangle + i\,|xz\rangle$ have the same energy—the full SU(3) symmetry is realized.

The eight gluons would then correspond to the eight independent ways of unitarily mixing the three orientation amplitudes: six off-diagonal generators (complex rotations between pairs of planes) and two diagonal generators (relative phase differences). This geometric identification parallels the proposal of Marsch and Narita [3], who connect the three spin-helicity degrees of freedom in 3D to SU(3) color structure. However, *demonstrating that the nonlinear dynamics of compressed Aether bridges respects the full SU(3) rather than only the SO(3) subgroup remains an open problem*—it requires showing that all eight collective excitation modes exist and that the bridge Lagrangian is invariant under arbitrary unitary mixing of orientations.

1.7 The Deconfinement Phase Transition

At extreme temperatures ($T \gtrsim 2 \times 10^{12}$ K, corresponding to ~170 MeV), hadronic matter undergoes a phase transition to a quark–gluon plasma (QGP), in which quarks and gluons move freely without confinement. This transition has been observed at the Relativistic Heavy Ion Collider (RHIC) and the Large Hadron Collider (LHC).

In the bridge model, the deconfinement transition is the *thermal disruption of the quark vortex endpoints that anchor the compressed bridges*. When thermal energy exceeds the anchoring energy per unit length, the quark vortices destabilize and the compressed strand relaxes back to ambient density. Without stable bridges, there is no confinement, and quark vortex endpoints move freely through the medium.

The bridge material itself remains in the same thermodynamic phase as the surrounding superfluid throughout this process (Section 4.8)—it is simply compressed by the topological boundary conditions at the quark endpoints. The deconfinement temperature $T_{\text{deconf}} \approx 170$ MeV sets the energy scale of the binding:

$$k_B T_{\text{deconf}} \approx \sigma \cdot d_{\text{bridge}} \tag{5}$$

where d_{bridge} is the effective bridge diameter. With $\sigma \approx 1$ GeV/fm, this gives $d_{\text{bridge}} \approx 0.17$ fm—the thermal energy scale of the bridge cross-section. This is smaller than the geometric bridge diameter (~0.3–0.35 fm from the lattice flux tube profile, Section 4.8), because the deconfinement condition probes the compact core of the bridge where the compression energy is concentrated, not the full width including the healing-length tails (~0.4–0.6 fm).

The QGP state is therefore the Aether equivalent of a high-temperature plasma: the compressed bridges have relaxed, and the formerly confined vortex endpoints (quarks) move through a hot medium. As the plasma cools below T_{deconf}, quark vortices restabilize, bridges re-form, and quarks are re-confined.

1.8 Quantum Fluctuations of the Bridge

The bridge model developed so far is classical: a static compressed strand connecting quark vortex endpoints. A complete description requires quantizing the fluctuations around this classical solution. The natural framework is *Bogoliubov–de Gennes theory*—the same linearization used in Section 4 for the homogeneous condensate, applied here to the cylindrical bridge geometry.

The bridge supports three fluctuation families: *longitudinal phonons* (compression waves at $c_{\text{phonon}} = c/\sqrt{X} \approx 10^{-4}\, c$, slow because K_A is constant while density is $X\rho_A$), *transverse oscillations* (string vibrations at speed c, since the mass per unit length equals σ/c^2), and *breathing modes* (gapped radial oscillations, frozen out at low energy). The transverse modes dominate the long-distance quantum correction.

The Lüscher term

The zero-point energy of the transverse fluctuation modes produces a quantum correction to the quark–antiquark potential. For a bridge of length L with quarks at its endpoints (Dirichlet boundary conditions), the allowed mode wavenumbers are $k_n = n\pi/L$. The zero-point energy per transverse direction is

$$E_{\mathrm{ZPE}} = \sum_{n=1}^{\infty} \frac{1}{2}\hbar\,\omega_n = \frac{\hbar\,c_s\,\pi}{2L}\sum_{n=1}^{\infty} n \tag{6}$$

The divergent sum is regulated by zeta-function regularization, $\sum n = \zeta(-1) = -1/12$, giving a finite contribution per polarization:

$$E_{\mathrm{ZPE}} = -\frac{\pi\,\hbar\,c_s}{24\,L} \tag{7}$$

In three spatial dimensions the bridge has two transverse polarization directions, so the total quantum correction to the static potential is

$$\Delta V(L) = -\frac{\pi\,\hbar\,c_s}{12\,L} \tag{8}$$

This is the bridge model's prediction of the *Lüscher term* [8, 9]. In QCD, the Lüscher term $\Delta V = -\pi/(12\,r)$ (in natural units with $c = 1$) is a universal quantum correction to the linear confining potential, confirmed by high-precision lattice simulations [10]. The bridge model reproduces the $-1/(12\,L)$ structure from the Bogoliubov zero-point energy of a finite-length condensate strand. The coefficient depends on the propagation speed of the transverse modes. The *longitudinal* (compression) modes propagate at $c_{\mathrm{phonon}} = \sqrt{K_A/\rho} = c/\sqrt{X} \approx 10^{-4}\,c$ inside the bridge, because the bulk modulus K_A is constant while the density is $X\rho_A$. However, the Lüscher term arises from *transverse* fluctuations, which in the Aether model are spin-wave excitations of the condensate—the same mode that constitutes light (Section 4). The spin stiffness K_{spin} governing these transverse modes scales with density ($K_{\mathrm{spin}} \propto \rho$, reflecting the stronger exchange coupling at closer particle spacing), giving a spin-wave speed $c_{\mathrm{spin}} = \sqrt{K_{\mathrm{spin}}/\rho} = c$ independent of compression. This is the standard condensed matter distinction between compression and spin-wave elastic moduli: the bulk modulus K_A (constant) governs longitudinal modes, while the spin stiffness $K_{\mathrm{spin}} \propto \rho$ governs transverse modes. The Lüscher coefficient is therefore set by $c_{\mathrm{spin}} = c$, matching the lattice-measured value exactly.

String width broadening

The transverse fluctuations cause the effective string width to grow logarithmically with quark separation:

$$w^2(L) = w_0^2 + \frac{\hbar\,c}{\pi\sigma}\,\ln\!\left(\frac{L}{L_0}\right) \tag{9}$$

where $w_0 \sim \xi$ is the intrinsic bridge width, L_0 is a short-distance cutoff of order ξ, and the coefficient $\hbar c/(\pi\sigma)$ sums over both transverse directions in 3+1 dimensions. This logarithmic broadening is a well-known prediction of the effective string theory of confinement [9], and lattice simulations confirm $w^2 \propto \ln L$ at large separations [2, 6]. At short separations ($L \lesssim 0.6$ fm), the measured width increases above the asymptotic trend [6], consistent with the bridge model's prediction that asymptotic freedom dissolves the compressed structure when $L \lesssim d_{\text{bridge}}$.

Virtual pair creation and the pion cloud

The most physically consequential fluctuation is *virtual bridge snapping*: a quantum fluctuation large enough to momentarily create a quark–antiquark pair along the bridge, producing a short-lived meson that is reabsorbed. The rate of such events is governed by the Schwinger mechanism [1] applied to the bridge's chromoelectric field. For a quark of mass m_q in a field of string tension σ, the pair-creation rate per unit length is

$$\frac{\Gamma}{L} \propto \sigma \, \exp\left(-\frac{\pi\, m_q^2\, c^3}{\hbar\, \sigma}\right) \tag{10}$$

The exponent evaluates to $\pi m_q^2 c^3/(\hbar\sigma) \approx \pi m_q^2/(\sigma \cdot \hbar c)$. Using $\hbar c \approx 197$ MeV fm and $\sigma \approx 1$ GeV/fm = 1000 MeV/fm:

Quark	m_q (MeV)	$\pi m_q^2/(\sigma\hbar c)$	Suppression
u, d	3–5	$\sim 10^{-4}$	None (copious)
s	~95	~0.14	Mild
c	~1275	~26	Strong
b	~4180	~280	Exponential

For light quarks (u, d), the exponential is essentially unity—pair creation is *unsuppressed*. The bridge is constantly dressed by a cloud of virtual pions (light $q\bar{q}$ bridge segments that briefly form and recombine). For charm and heavier quarks, pair creation is exponentially suppressed, and the bridge is a clean, well-defined string.

The crossover mass where pair creation turns off (Schwinger exponent ~ 1) is $m_q^* \sim \sqrt{\sigma\,\hbar c/\pi} \approx \sqrt{63{,}000}$ MeV ≈ 250 MeV—between the strange and charm quark masses. This reproduces a fundamental feature of QCD: the transition from light-quark (chiral) physics, dominated by pion-cloud effects, to heavy-quark (potential model) physics, where the static string picture is accurate. In the bridge model, this transition is not a change in theoretical framework but a single physical parameter—the Schwinger exponent—passing through unity.

Implications for the nucleon sigma term

The pion cloud has a direct physical consequence: the nucleon sigma term $\sigma_{\pi N} = \hat{m}\,\langle N|\bar{u}u + \bar{d}d|N\rangle$, where $\hat{m} = (m_u + m_d)/2$. This measures how much of the nucleon mass depends on the light quark masses. In the classical bridge model (Section 1), the quarks are ultra-relativistic: their momentum ($p \sim \hbar/L_{\text{arm}} \approx 660$ MeV/c) vastly exceeds their rest mass ($m_q c \approx 5$ MeV/c), so the bridge equilibrium geometry is insensitive to quark mass changes. The Hellmann–Feynman theorem gives a classical sigma term of only $\sigma_{\pi N}^{\text{classical}} \approx 3\hat{m}^2 c^2/(pc) \sim 0.06$ MeV—essentially zero.

The measured value, $\sigma_{\pi N} \approx 45$–$60$ MeV [11, 12], is overwhelmingly dominated by virtual pion exchange: the bridge fluctuations create and absorb virtual pions whose mass depends on m_q through the Gell-Mann–Oakes–Renner relation ($m_\pi^2 \propto \hat{m}$). The bridge model thus predicts the correct hierarchy: a negligible *classical* sigma term (from static geometry) plus a large *quantum* contribution (from the pion cloud dressed by Schwinger pair creation), with the quantum piece dominating by two orders of magnitude.

Computing the full $\sigma_{\pi N}$ from the bridge model requires calculating the pion–nucleon coupling constant $g_{\pi N}$ from the overlap of the virtual pion's bridge wavefunction with the Y-junction geometry of the nucleon—an open calculation that would provide a stringent quantitative test.

1.9 The Weak Force: Vortex Topology Transitions

The weak nuclear force is responsible for processes that change one type of quark or lepton into another—most notably beta decay, where a neutron transforms into a proton, an electron, and an antineutrino. In the Standard Model, the weak force is mediated by the massive $W^\pm$ and Z^0 bosons and is described by the SU(2) gauge symmetry.

In the Aether framework, the weak force is fundamentally different from the other three: it is not a force in the traditional sense but a *topology-changing transition* of the vortex structure. When a down quark transforms into an up quark (emitting a W^- boson), the vortex endpoint changes its circulation configuration—the bridge attached to it restructures.

The SU(2) gauge symmetry of the weak force arises from the spinor order parameter $\Psi = (\psi_\uparrow, \psi_\downarrow)$ that is already required by the theory for spin-statistics. The W and Z bosons are massive because they correspond to *gapped* excitations of the spinor condensate: changing the relative amplitude or phase of the two spinor components costs energy proportional to the condensate stiffness. Their large masses ($m_W \approx 80.4$ GeV, $m_Z \approx 91.2$ GeV) reflect the enormous energy required to restructure the condensate at the scale of a single vortex.

The short range of the weak force ($\sim 10^{-18}$ m) follows from the large boson masses through the uncertainty principle: a virtual W or Z can propagate only a distance $\sim \hbar/(m_W c) \approx 10^{-18}$ m before being reabsorbed. In Aether terms, the topology-changing disturbance is heavily damped because it requires restructuring the dense condensate, and it cannot propagate far before the condensate reasserts its ground-state configuration.

Parity violation—the weak force's coupling only to left-handed particles—may reflect a fundamental chirality in the vortex topology. If the binding bridge attaches preferentially to one circulation handedness of the vortex endpoint, the topology-changing transitions mediated by W bosons would naturally couple to only one chirality. This remains an open question requiring detailed analysis of vortex–bridge junction geometry.

1.10 The Four Forces Unified Through Aether Mechanics

The four fundamental forces now have a unified interpretation within the Aether framework:

Force	**Mechanism**	**Range**	**Strength**
Strong	Compressed Aether bridge	$\sim$1 fm (bridge snaps)	$\sigma \approx 1$ GeV/fm
EM	Flow vector of unbound Aether	Infinite ($1/r^2$)	$\alpha \approx 1/137$
Weak	Vortex topology transition	$\sim 10^{-18}$ m	$G_F \sim 10^{-5}$ GeV^{-2}
Gravity	Bind–unbind cycle asymmetry	Infinite ($1/r^2$)	$\alpha_G \sim 10^{-39}$

The force hierarchy is natural:

- The strong force is the strongest because it involves the *full binding energy* of condensed Aether—the most energetic process available in the medium.
- The electromagnetic force is weaker because it involves the *pressure of flowing* unbound Aether—significant, but not binding-scale.
- The weak force is weaker still because it requires *restructuring the condensate*—a costly process damped by the large condensate stiffness.
- Gravity is the weakest because it is only the *tiny residual asymmetry* of the transient bind–unbind cycle—the faintest whisper of the binding process.

All four forces arise from the same medium and its mechanical capabilities. The strong force is elastic compression anchored by quark vortex endpoints. Electromagnetism is the flow pattern of unbound Aether. The weak force is the topological restructuring of vortex endpoints. Gravity is the leftover asymmetry of transient binding. There is no need for separate, unrelated mechanisms—the four forces are four aspects of one medium.

2. Summary of Theoretical Properties

2.1 Properties of the Strong Force

1. Quark confinement arises from compressed Aether bridges—condensate strands connecting quark vortex endpoints, with string tension $\sigma \approx 1$ GeV/fm set by the elastic compression energy of the logarithmic equation of state.

2. Bridges are elastic compressions of the same superfluid phase (compression ratio $X \approx 7 \times 10^7$), not a separate bound phase. Deconfinement at $T \approx 170$ MeV corresponds to thermal disruption of the anchoring quark vortices, not melting of a distinct phase.

3. Asymptotic freedom follows from bridge compactness: $d_{\text{bridge}}/\xi \approx 1$–2 means bridges are "all core" with no bulk interior. At separations $L < d_{\text{bridge}}$, the condensate perturbation has not fully formed, giving weak coupling; at $L > d_{\text{bridge}}$, full string tension emerges. The Coulomb-to-linear crossover at $L \sim \xi \approx 0.2$ fm matches lattice QCD with no free parameters.

4. Color charge corresponds to quantum superpositions of three bridge orientations (xy, xz, yz planes); unitary mixing of these three complex amplitudes gives SU(3) symmetry with 8 generators. Full SU(3) invariance (beyond the SO(3) subgroup) requires the bridge dynamics to be invariant under relative complex-phase rotations between orientations.

5. Hadron mass is predominantly bridge energy ($\sigma \times L$), not quark rest mass: three bridge arms of length ~0.3 fm give $3\sigma L_{\text{arm}} \approx 900$ MeV, reproducing the proton mass with quark and junction corrections.

6. Quantum fluctuations of the bridge produce three mode families: longitudinal phonons ($c_{\text{phonon}} = c/\sqrt{X}$ at compression X, from the constant-K_A logarithmic EOS), transverse spin-wave oscillations ($c_{\text{spin}} = c$, from $K_{\text{spin}} \propto \rho$), and gapped breathing modes.

7. The Lüscher term $\Delta V = -\pi\hbar c_s/(12L)$ emerges from the zero-point energy of two transverse polarizations, reproducing the universal $-1/(12\,r)$ quantum correction to the confining potential.

8. Schwinger pair creation along the bridge produces a pion cloud for light quarks (exponent $\ll 1$) but is exponentially suppressed for heavy quarks. The crossover mass $m_q^* \approx 250$ MeV naturally separates chiral QCD from potential-model QCD.

9. String-breaking at $L \approx 1.2$ fm (matching lattice QCD) occurs when the bridge stores enough energy for quark–antiquark pair creation including kinetic energy.

10. The flux tube width ($w \approx 0.5$ fm) matches the bridge model prediction 2–3 $\xi \approx 0.4$–0.6 fm with no adjustable parameters.

11. The weak force arises from vortex topology transitions (core reconnection of quark vortex endpoints), with the short range ($\sim 10^{-18}$ m) set by the massive W and Z bosons as gapped excitations of the spinor condensate.

3. Experimental Support and Connections to Established Physics

The following experimentally confirmed phenomena support or align with the aspects of the theory presented in this paper:

1. **Lattice QCD string-breaking distance.** Lattice simulations with dynamical quarks [4, 5] observe the static quark–antiquark potential levelling off at a separation $r_{\text{break}} \approx 1.2$ fm, confirming string breaking by light-quark pair creation. The bridge model predicts this distance from the string tension $\sigma \approx 1$ GeV/fm and the pair-creation energy threshold (~1 GeV, including rest mass and kinetic energy of the created quarks), giving $L_{\text{snap}} \approx 1.2$ fm—in direct quantitative agreement with the lattice measurement.

2. **Lattice QCD flux tube width.** High-precision lattice simulations [2, 6] measure the transverse chromoelectric field width at $w \approx 0.5$ fm for quark separations $d \gtrsim 0.7$ fm. The bridge model predicts the compressed region spans ~ 2–$3\,\xi \approx 0.4$–0.6 fm, in quantitative agreement with no adjustable parameters. Earlier dual-superconductor fits [7] extract a penetration depth $\lambda \approx 0.22$–0.24 fm that overlaps with the Aether healing length $\xi \approx 0.17$–0.28 fm.

3. **Lüscher term in the static quark potential.** Lattice QCD simulations [10, 9] confirm that the quark–antiquark potential at large separations deviates from pure linearity by a universal quantum correction $\Delta V = -\pi/(12\,r)$. The bridge model derives this term from the zero-point energy of transverse Bogoliubov modes confined to a finite-length condensate strand (Section 1.8), reproducing the $-1/(12\,r)$ structure with the coefficient set by the transverse (spin-wave) speed $c_{\text{spin}} = c$, which is density-independent because the spin stiffness scales linearly with density ($K_{\text{spin}} \propto \rho$).

4. **Light-quark versus heavy-quark phenomenology.** QCD exhibits a sharp transition: light quarks (u, d, s) are governed by chiral dynamics and strong pion-cloud effects, while heavy quarks (c, b) are accurately described by static potential models. The bridge model reproduces this transition from a single mechanism: the Schwinger pair-creation exponent $\pi m_q^2/(\sigma\hbar c)$ passes through unity at $m_q^* \approx 250$ MeV, between the strange and charm masses, naturally separating copious pion-cloud dressing (light quarks) from clean string dynamics (heavy quarks).

5. **Asymptotic freedom.** Deep inelastic scattering experiments confirm that quarks interact more weakly at short distances. In the bridge model this is geometric: when quarks are close, the

compressed Aether bridge between them is short and nearly all core ($d/\xi \approx 1$–2), so there is little compressed material to resist further approach.

6. **Hadron mass from bridge energy.** The proton mass (≈ 938 MeV) far exceeds the summed current quark masses (≈ 10 MeV). The bridge model attributes the excess to compressive energy stored in the Aether bridges: string tension times bridge length ($\sigma \times L \approx 900$ MeV) reproduces the measured mass with quark and junction corrections.

7. **The deconfinement phase transition.** Heavy-ion collisions at RHIC and the LHC produce quark–gluon plasma at $T \approx 170$ MeV, in which quarks move freely. In the bridge model, this is thermal disruption of compressed bridges: when thermal energy exceeds the bridge binding energy, the compressed Aether channels dissolve and quarks deconfine.

4. Open Questions and Testable Predictions

The following areas relevant to this paper remain underdeveloped and represent opportunities for further refinement or falsification:

1. **Nucleon sigma term from bridge fluctuations.** The classical bridge model predicts a negligible sigma term ($\sigma_{\pi N}^{\text{classical}} \sim 0.1$ MeV) because the quarks are ultra-relativistic and the bridge geometry is insensitive to their mass. The measured $\sigma_{\pi N} \approx 45$–60 MeV [11, 12] is dominated by the pion cloud arising from Schwinger pair creation along the bridge (Section 1.8). Computing the full sigma term requires deriving the pion–nucleon coupling $g_{\pi N}$ from the overlap of the virtual-pion bridge wavefunction with the Y-junction geometry—a calculation that would test the quantum bridge framework against one of the most precisely known hadronic quantities.

5. Mathematical Summary

This section collects the key equations relevant to the topics covered in this paper. *For the complete mathematical summary, see Paper 1: General Overview [13].*

5.1 Fundamental Aether Parameters

The theory posits four microscopic parameters from which all macroscopic phenomena derive:

Symbol	Name	Constrained range	Primary constraint
ρ_A	Ambient density	10^{11}–10^{12} kg/m^3	Electron vortex energy
K_A	Bulk modulus	10^{28}–10^{29} J/m^3	$K_A = \rho_A c^2$
m_A	Aether quantum mass	500–850 MeV/c^2	Flux tube width / glueball mass
ξ	Healing length	0.17–0.28 fm	Bridge diameter (lattice QCD)

The Coulomb-constant cross-check of Paper 7(a) [14], Section "Tightened Parameter Ranges," narrows these ranges further: $m_A \in [628, 689]$ *MeV/c^2,* $\rho_A \in [8.9 \times 10^{11}, 1.07 \times 10^{12}]$ *kg/m^3,* $\xi \in [0.20, 0.22]$ *fm,* $K_A \in [8.0 \times 10^{28}, 9.6 \times 10^{28}]$ *J/m^3. The compression ratio X along the Coulomb curve is* $\approx 5.2 \times 10^7$ $(\log_{10} X \approx 7.71)$*, about* 35% *below the electron-mass-only value* $\approx 7 \times 10^7$ *quoted in this paper; the difference reflects the* f_{core}*-reconciliation caveat discussed in Paper 7(a).*

These four are not independent. Two defining relations connect them:

$$c = \sqrt{K_A/\rho_A} \tag{11}$$

$$\xi = \frac{\hbar}{m_A\, c\,\sqrt{2}} \tag{12}$$

leaving two free Aether parameters (e.g., ρ_A and m_A) from which all others follow. (A third parameter, ϵ, enters through gravity; see below.)

5.2 Equation of State

The Aether obeys a logarithmic equation of state, the unique form that produces a density-independent bulk modulus (Section 4):

$$P = K_A \ln\!\left(\frac{\rho}{\rho_0}\right) + P_0 \tag{13}$$

Key consequence: the speed of sound $c_s = \sqrt{K_A/\rho}$ varies with density. At ambient density, $c_s = c$. Note: in gravitational fields, Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly, so c does not vary near gravitating masses; the density dependence is relevant for material media and for the two-fluid cosmological structure. This equation of state guarantees that K_A is constant under compression, which is experimentally confirmed by:

- Velocity time dilation measurements spanning $10^{-6}\, c$ to $0.9994\, c$ (precision $\sim 10^{-8}$).
- The Fresnel drag coefficient $1 - 1/n^2$, exact across materials with density ratios up to ~6:1.

5.3 Strong Force (Bound Aether Bridges)

The strong force is a physical strand of compressed Aether connecting quark vortex endpoints (Section 1).

String tension.

$$\sigma \approx 1\ \text{GeV/fm} = 1.6 \times 10^{5}\ \text{J/m} \tag{14}$$

Bridge diameter from deconfinement.

$$d_{\text{bridge}} = \frac{k_B\, T_{\text{deconf}}}{\sigma} \approx 0.17\ \text{fm} \tag{15}$$

Bridge energy density.

$$\varepsilon_{\text{bridge}} = \frac{\sigma}{A} = \frac{4\sigma}{\pi\, \xi^2} \sim 10^{36}\text{–}10^{37}\ \text{J/m}^3 \tag{16}$$

Compression ratio. The ratio of bridge to ambient density is determined by the three constrained quantities σ, ξ, and K_A (Section 4.8):

$$X \equiv \frac{\rho_{\text{bridge}}}{\rho_A} = \frac{4\sigma}{\pi\, \xi^2\, K_A} \approx 7 \times 10^{7} \tag{17}$$

The compression energy from the logarithmic equation of state:

$$\varepsilon_{\text{compress}} = K_A \left[X - \ln X - 1\right] \approx K_A\, X \quad (\text{for } X \gg 1) \tag{18}$$

Proton mass from bridge geometry. Three Y-junction arms of length ~0.3 fm:

$$m_p\, c^2 \approx 3\, \sigma\, L_{\text{arm}} + m_{\text{quarks}}\, c^2 + E_{\text{junction}} \approx 900 + 9 + 29 = 938\ \text{MeV} \tag{19}$$

where $E_{\text{junction}} \approx 29$ MeV is the residual binding energy at the Y-shaped junction, inferred from the difference between the measured proton mass and the bridge plus quark contributions.

Wave speeds in bridge. The bridge has two distinct propagation speeds:

$$c_{\text{phonon}} = \frac{c}{\sqrt{X}} \approx 10^{-4}\, c \approx 3 \times 10^{4}\ \text{m/s} \quad (\text{longitudinal, from constant } K_A) \tag{20}$$

$$c_{\text{spin}} = c \quad (\text{transverse, from } K_{\text{spin}} \propto \rho) \tag{21}$$

The Lüscher term coefficient is set by $c_{\text{spin}} = c$, matching the lattice QCD value $\Delta V = -\pi/(12\, r)$.

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